

The Kinetic Mechanism of Ubiquinol: Cytochrome *c* Reductase at Steady State

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The steady-state kinetics of ubiquinol: cytochrome *c* reductase (cytochrome *bc*₁ complex) is analyzed in this work. The graphical pattern of the titrations is clearly indicative of a ping-pong mechanism, but the two products ubiquinone and reduced cytochrome *c* behave competitively with their substrate and noncompetitively with the other substrate. Hence, the mechanism of the reductase is of a ping-pong two-site type. A minimal reaction scheme for the enzymatic mechanism is proposed and approximate values of its rate constants are deduced on the assumption that each substrate is in rapid equilibrium at its catalytic site. This has been substantiated by pre-steady-state measurements of the reduction and oxidation of cytochrome *b* by a short-chain homolog of ubiquinol. Values of the rate constants of the reaction scheme have been deduced from the steady-state titrations for a series of 2,3-dimethoxy-5-methyl quinols having different hydrophobic substituents in position 6 of the ring. The results provide a quantitative estimation of the specificity of the quinol catalytic site in the transmembrane portion of the *bc*₁ complex. In particular, a reasonable correlation is found between the rate of the second-order reaction of quinols with the enzyme and their solubility in lipids.

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The ubiquinol:cytochrome *c* reductase (EC 1.10.2.2, also called *bc*₁ complex or Complex III; Ref. (1)) constitutes the central enzymatic complex of the mitochondrial respiratory chain. This enzyme oxidizes the ubiquinol which is formed by the various dehydrogenases (e.g., NADH dehydrogenase) via a reaction switch from a hydrogen carrier (ubiquinol) to single electron carriers (cytochromes). The four prosthetic groups of the *bc*₁ complex are physically located into two domains (1, 2): (i) the membrane domain which communicates with ubiquinol and includes the di-

haem cytochrome *b*, and (ii) the extrinsic domain that interacts with cytochrome *c* at the positive side of the membrane and includes the "Rieske" 2Fe2S cluster and cytochrome *c*₁.

The catalytic mechanism of the *bc*₁ complex has been mainly studied by following the transient redox kinetics of its prosthetic groups (3–10). Modified versions of the Q-cycle scheme first introduced by P. Mitchell (11) represent the best descriptions of the observed redox kinetics in the enzyme (2, 4, 6, 12–14). Specific inhibitors of the *bc*₁ complex such as antimycin and myxothiazol have been essential for the development of these cyclic schemes and the design of the pre-steady-state measurements which support them (3–16). However, the use of inhibitors generates the suspicion that they may alter some aspects of the catalytic steps characteristic of the native enzyme (1, 9, 17, 18).

Steady-state measurements of the enzymatic activity of the *bc*₁ complex can provide valuable information on the reaction mechanism of the native enzyme (19, 20). However, the enzymatic assay of the *bc*₁ complex is complicated by a series of problems which have hampered, so far, a complete characterization of the reductase at steady state. The main problems derive from the substrate inhibition exerted by high concentrations of cytochrome *c* (19–22), the nonlinear saturation kinetics that sometimes is displayed by ubiquinols at low cytochrome *c* concentrations (19, 23), and the poor solubility of quinol homologs in water (20, 24, 25).

The understanding of the interaction of ubiquinols with their oxidizing enzymes requires a detailed description of their pre-steady-state and steady-state kinetic properties. Toward this end, we have recently elucidated the mechanism of ubiquinol oxidation by the isolated Rieske protein (26) and analyze here the steady-state kinetics of the native *bc*₁ complex in great detail.

MATERIALS AND METHODS

Materials. Mitochondrial *bc*₁ complex from beef heart, chicken heart, and turkey heart was purified as described previously (22). The beef

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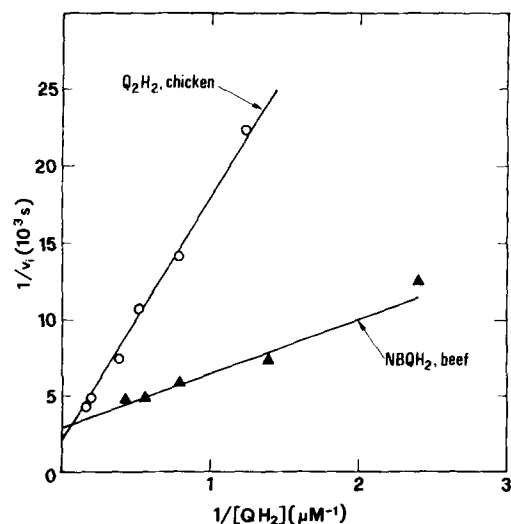


FIG. 1. Reductase titrations at a fixed ratio of the two substrates. The chicken bc_1 complex was titrated with a fixed ratio of 1.15 between the concentration of Q_2H_2 and that of cytochrome c , whereas the beef bc_1 complex was titrated with a fixed ratio of 5.17 between the concentration of $NBQH_2$ and that of cytochrome c . Similar linear plots are obtained in other enzymes and/or with other quinols (52). The initial rates of the enzymatic activity (v_i) are expressed in turnover numbers, i.e., moles of cytochrome c reduced per mole of cytochrome c_1 per second.

heart complex was purified also with the method of Engel *et al.* (27). Samples of bc_1 complex purified from baker's yeast were kindly provided by Dr. A. L. Tsai and Professor G. Palmer (Rice University, Houston).

Ubiquinol were prepared from ubiquinones by the method of Rieske (28) and suspended at 2–16 mM in acidified ethanol. Ubiquinone homologs were kindly offered by Eisai Co. (Tokyo, Japan). Pentyl-benzoquinone (PBQ)² was given by Professor K. Folkers (University of Texas, Austin), whereas decyl-benzoquinone (DBQ) was purchased from Professor B. L. Trumpower (Dartmouth College, Hanover, NH). Nonyl-benzoquinone (NBQ) was kindly provided by Professor G. Von Jagow (Goethe-Universität, Frankfurt-am-Main, Germany), whereas 2,3-dimethoxy-5,6-dimethyl-benzoquinone (DDQ) was offered by Professor A. Kröger of the same university. Duroquinol (2,3,4,5-tetramethylbenzoquinol, abbreviated here as DQH_2) was purchased from K&K Labs.

Determinations. Cytochrome c (horse, type VI Sigma) was measured at 550–540 nm after reduction with an extinction coefficient of 19 mM^{-1} (19). Cytochrome c_1 was determined after ascorbate reduction of the bc_1 complex at 552–541 nm using an extinction coefficient of 17.5 mM^{-1} (29). Measurements of cytochrome b reduction and oxidation under non-steady-state conditions were performed as previously described (9).

The midpoint redox potential of cytochrome c_1 was measured spectrophotometrically (30) and that of commercial cytochrome c was evaluated under the same conditions as those of the reductase assay by either MCD measurements as described in Ref. (31) or by chemical equilibration with ferricyanide (32).

Methods. The bc_1 preparations were diluted to ca. $1 \text{ } \mu\text{M}$ in a 20 mM Tris-Cl buffer, pH 7.4, containing 0.2% Tween-20 and 0.5% bovine serum

albumin and then used at 2–5 nM in the assay cuvette containing variable concentrations of cytochrome c (usually not exceeding $10 \text{ } \mu\text{M}$ to minimize substrate inhibition; Refs. (20, 22)) in 25 mM K-phosphate buffer, pH 7.4, 1 mM EDTA and 1 mM KCN. This buffer is not at the pH optimum of the reductase, which is around 8 (20, 26), but offers a reasonable compromise for reducing the extent of chemical reduction of cytochrome c by the quinols. Contrary to previous works (20, 28), the addition of albumin to the assay buffer has been avoided, for this reagent affects the binding of the quinols (M. Degli Esposti and M. Barilli, unpublished results).

The reaction was started with the quinol substrate (generally at concentrations > one-half of its usual K_m for avoiding too much scatter in the data and/or nonlinear behaviors of the rates cf. Ref. (19)) and monitored at room temperature as described (22, 26). The chemical reaction was usually measured with the same enzyme concentration in the presence of an excess of both antimycin and myxothiazol.

The binding of cytochrome c to oxidized beef heart bc_1 complex was performed as described in Ref. (20) under the same conditions of the reductase assay. The Scatchard plots could be fitted by a single binding site with $K_d = 0.8 \text{ } \mu\text{M}$ (mean of three determinations) for oxidized cytochrome c .

The binding of exogenous ubiquinones to the beef heart bc_1 complex was measured by following the quenching of the intrinsic tryptophan fluorescence (33).

RESULTS

1. The Steady-State Behavior of the Ubiquinol: Cytochrome c Reductase Conforms to a Ping-Pong Two-Site Mechanism

Figure 1 shows that the rates of the reductase activity of bc_1 complexes purified from two different species are linearly dependent upon the substrates' concentration held at a fixed ratio. This linear behavior is a strong indication for a ping-pong mechanism (34). Indeed, the titrations as a function of the cytochrome c substrate consistently show a family of parallel lines (Fig. 2A), a behavior which is also suggestive of a ping-pong mechanism (10, 20, 34). However, the lines obtained in the titrations of the ubiquinol substrate often tend to intersect each other (19, 22). A careful evaluation of the quinol titrations indicates that even in these cases the lines are essentially parallel to each other when the concentration of cytochrome c is below 6–10 μM . High concentrations of cytochrome c induce an increase in the slope and a decrease of the intercept of the quinol titrations (Fig. 2B, see also Ref. (22)), thus showing a (partial) competitive substrate inhibition (34).

The type of product inhibition displayed by both oxidized quinones and reduced cytochrome c , however, is not that expected for a typical ping-pong mechanism (Fig. 3 and Ref. (20)). Both products behave as competitive inhibitors of their respective substrates (e.g., ubiquinone vs ubiquinol; Fig. 3A), whereas they are expected to behave noncompetitively since they should bind to different enzymatic forms in a ping-pong mechanism (34–36). Such an unusual behavior is seen in other enzymes that also present the graphical pattern characteristic of a ping-pong scheme (36–40). The most straightforward interpretation of this kinetic phenomenon is that ubiquinol

² Abbreviations used: cyt, cytochrome; $DBQH_2$, 2,3-dimethoxy,5-methyl-6-decyl benzoquinol; DQH_2 , duroquinol; $DDQH_2$, 2,3-dimethoxy,5,6-dimethyl-benzoquinol (aurantocladiol); $NBQH_2$, 2,3-dimethoxy,5-methyl,6-nonyl benzoquinol; $PBQH_2$, 2,3-dimethoxy,5-methyl,6-pentyl benzoquinol; Q , ubiquinone; QH_2 , ubiquinol; TN, turnover number.

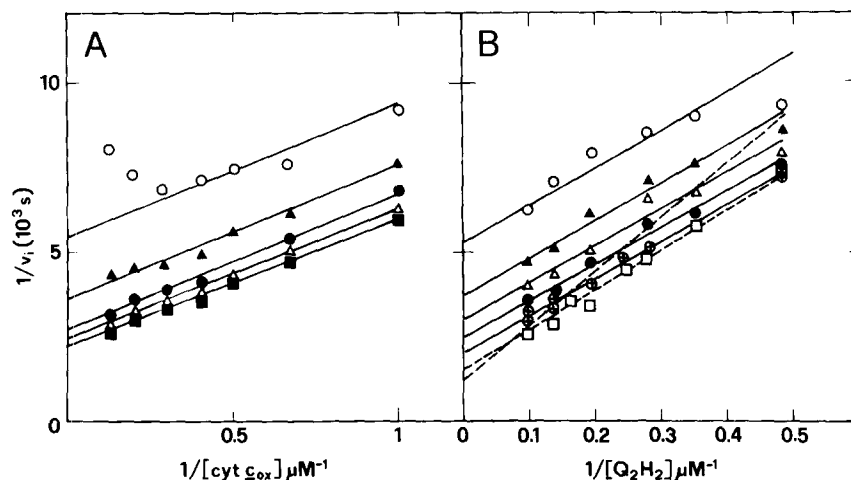


FIG. 2. Titrations of the ubiquinol-2:cytochrome *c* reductase in the beef heart bc_1 complex. (A) Cytochrome *c* titrations at the following fixed concentrations of Q_2H_2 : (○) 2.1 μM ; (▲) 3.6 μM ; (●) 5.2 μM ; (△) 7.3 μM ; (■) 10.4 μM . The data points obtained at 20 μM quinol are nearly superimposable to those obtained at 10.4 μM quinol (results not shown for sake of clarity). (B) Ubiquinol titrations at the following fixed concentrations of cytochrome *c*: (○) 1 μM ; (▲) 1.5 μM ; (△) 2 μM ; (●) 2.5 μM ; (⊕) 5 μM ; (□) 8 μM . Note the increase in the slope of the last curve. Such an increase becomes more and more pronounced by further increasing the concentration of cytochrome *c* (22), as shown by the short-dashed curve that corresponds to the titration obtained at 20 μM of this substrate (its data points are omitted for sake of clarity).

and cytochrome *c* interact at two distinct catalytic sites in the enzyme (20, 36–40), a concept which is largely supported by the present knowledge of the structural organization of the bc_1 complex (1, 2, 9, 14, 20). Thus, the steady-state catalysis of the bc_1 complex essentially conforms to a ping-pong two-site mechanism, the min-

imal description of which (illustrated in Fig. 4) is identical to that of a one-site ping-pong mechanism if each substrate is assumed to be in rapid equilibrium at its catalytic site (37–40). Following this assumption, the maximal turnover (TN_{max}), $K_m QH_2$, and $K_m c$ obtained experimentally derive from the eight rate constants of

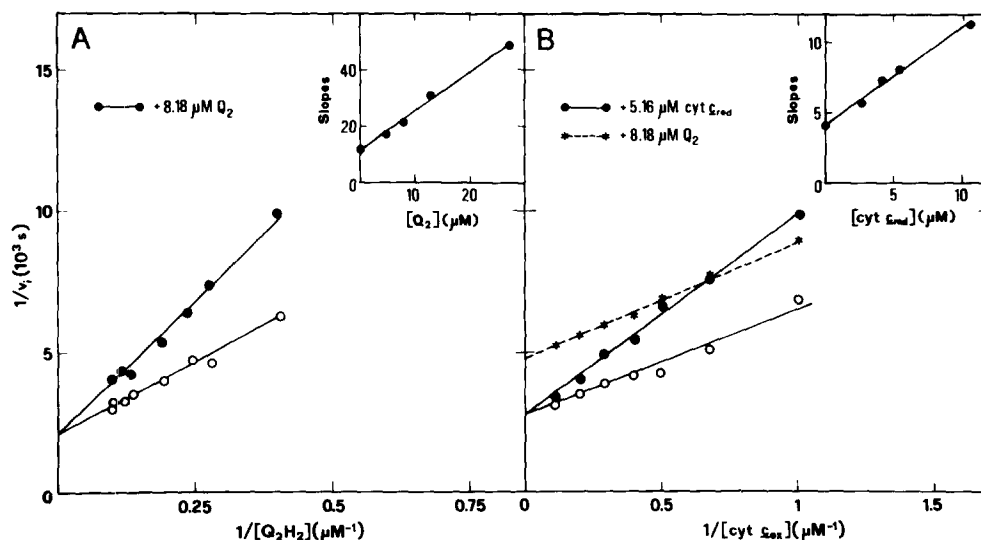


FIG. 3. Products' inhibition of the Q_2H_2 :cytochrome *c* reductase in the beef heart bc_1 complex. (A) Product inhibition of Q_2 versus Q_2H_2 at 5 μM cytochrome *c*; (○) control titration without Q_2 . The insert shows the Dixon plot of the slope effect of Q_2 , from which a $K_i Q$ of 8.9 μM is calculated according to Ref. (36). (B) Product inhibition of reduced cytochrome *c* ($cyt c_{red}$) and of Q_2 versus oxidized cytochrome *c* at 5.2 μM Q_2H_2 ; (○) control titration without either product. The insert shows the Dixon plot of the slope effect of $cyt c_{red}$, from which a $K_i c_{red}$ of 5.2 μM is calculated according to Ref. (36). When this value is averaged with those obtained in three other separate experiments and with that in Ref. (20), a mean $K_i c_{red}$ of 4 μM is obtained. Similarly to Q_2 versus cytochrome *c*, reduced cytochrome *c* also produces an uncompetitive inhibition versus Q_2H_2 (52). All these patterns of products' inhibition conform to those predicted for a ping-pong two-site mechanism with rapid-equilibrium at both sites (36, 37, 40).

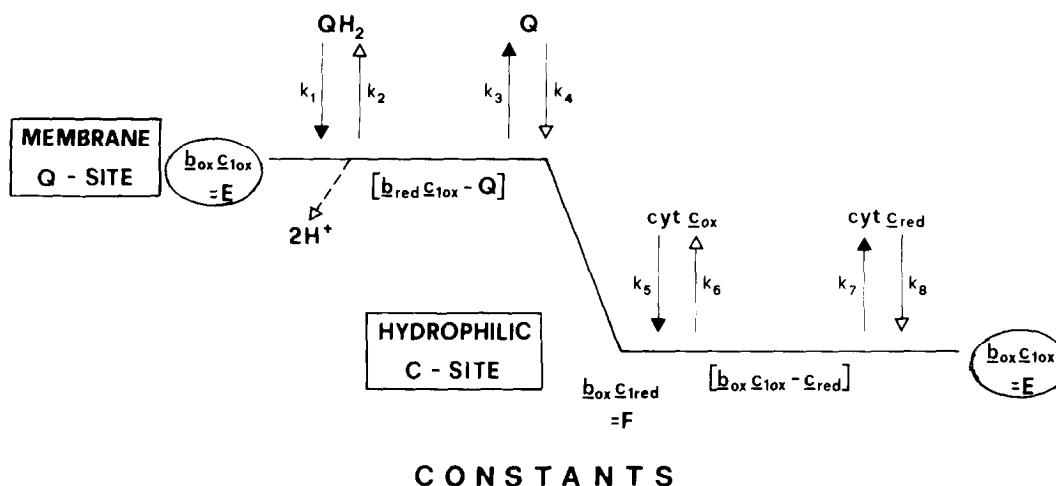


FIG. 4. Minimal reaction scheme for the steady-state mechanism of the ubiquinol:cytochrome *c* reductase. The scheme follows the conventions for a typical ping-pong two-site mechanism (34, 37, 40). It contains the minimal number of rate constants that can describe the relevant kinetic steps of addition and release of substrates and products at each site. The deprotonation of ubiquinol at the Q site is considered to be very fast with respect to the electron transfer reactions (2, 4, 10, 26). The principal enzyme-substrate complex at the Q site is represented as the enzymatic form in which one electron resides on cytochrome *b*, while the second electron from ubiquinol is assumed to reside (transiently) in the Rieske cluster. Note that one turnover of the Q site actually transfers one electron back to Q at center *i* (2, 6, 12), even if two electrons are taken from ubiquinol. See Refs. (6, 10, 12) for a complete evaluation of all the intermediate steps that occur during oxidation of exogenous ubiquinol by the *bc*₁ complex. Most of such steps are omitted here as they should not be rate-limiting in the assumption that the Q site is essentially in rapid equilibrium at steady state. Even if a definite distinction of center *o* and *i* is not possible in this simplified scheme, it is conceivable that *k*₃ and *k*₄ represent reactions that occur essentially at center *i* (see text and Refs. (9, 10, 12, 58)). The principal enzyme-substrate complex at the *c* site is considered to be the enzymatic form having one electron in cytochrome *c*₁ and with cytochrome *b* oxidized. This is a simplification to stress that the interaction of cytochrome *c* with its site is independent of the redox state of both cytochrome *b* and endogenous ubiquinone at the Q site (actually at center *i* at equilibrium (10, 12, 45)). Other likely intermediate complexes of the catalytic interaction with cytochrome *c*₁ (cf. Refs. (20, 44)) are omitted here under the assumption that the *c* site is in rapid equilibrium. This assumption is sustained by the pattern of product inhibition (Fig. 3B and Refs. (20, 37, 40, 52)) and by the properties of the redox reaction between cytochrome *c* and cytochrome *c*₁ Refs. (29, 41-44).

the scheme illustrated in Fig. 4, according to the simplified equations (34)

$$TN_{\max} = V_{\max}/[bc_1] = k_3 k_7 / (k_3 + k_7) \quad [1]$$

$$K_m QH_2 = k_7 \times (k_2 + k_3) / k_1 \times (k_3 + k_7) \quad [2]$$

$$K_m c = k_3 \times (k_6 + k_7) / k_5 \times (k_3 + k_7). \quad [3]$$

The ratio $TN_{\max}/K_m$ of each substrate ($k_{\min}$) expresses the minimal value of its second-order rate constant of reaction with its catalytic site (25, 34). The relationship of $k_{\min} QH_2$ and $k_{\min} c$ with the constants of the reaction scheme of the *bc*₁ complex (Fig. 4) is given by Eqs. [4] and [5], respectively:

$$k_{\min} QH_2 = k_1 \times (k_3/k_2 + k_3) \quad [4]$$

$$k_{\min} c = k_5 \times (k_7/k_6 + k_7). \quad [5]$$

The two catalytic sites of the reductase can be char-

acterized independently if conditions are selected in which the kinetic interaction of one substrate is overall rate-limiting.

2. Characterization of the Kinetic Interaction of Cytochrome *c*

The kinetic interaction of the cytochrome *c* substrate is usually rate-limiting when the reductase assay is performed under optimal assay conditions and by using quinols such as Q₂H₂ or DBQH₂ (20, 22). The interaction of the cytochrome *c* catalytic site in the extrinsic domain of the complex fundamentally involves an electrostatic binding of the cytochrome followed by a rapid electron transfer with cytochrome *c*₁, which is in very rapid redox equilibrium with the Rieske cluster (8, 9).

Isolated cytochrome *c*₁ reacts with cytochrome *c* at very fast rates, in a reaction that is essentially governed by the thermodynamic equilibrium of the two haemoproteins (41, 42). As the characteristics of the redox interaction

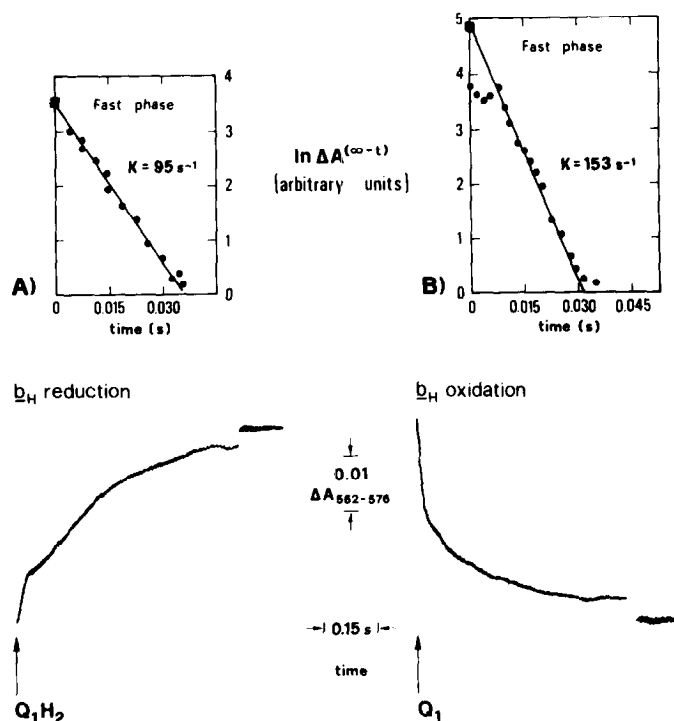


FIG. 5. Pre-steady-state reactions of ubiquinol-1 and ubiquinone-1 with cytochrome *b* in the partially reduced bc_1 complex from beef heart. The traces of the redox changes specific of cytochrome *b* are obtained in a stopped-flow apparatus having a dead plus mixing time of 4 ms (9) (the filled square in the insert of both A and B represents the absorbance at time = 0) in the same buffer of the enzymatic assays. The final concentration of the bc_1 complex is 2.9 μM (5.5 μM cytochrome *b*). (A) Reduction of cytochrome *b* by 49 μM Q_1H_2 in the ascorbate-reduced complex, where only the Rieske cluster and cytochrome c_1 are reduced (1, 8, 9). The fast phase of the biphasic reaction is shown in the insert and corresponds to a second-order rate constant of $1.94 \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$. This value is similar to the average rate constant measured in the fully oxidized complex (ca. $3 \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$ in several separate experiments), in agreement with Refs. (9, 10, 57). (B) Oxidation of cytochrome *b* pre-reduced by 2 mM succinate followed by 5 mM malonate (9) with 59 μM Q_1 . The fast phase of the biphasic reaction, shown in the insert, corresponds to a second-order rate constant of $2.59 \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$. The rate constant of the reaction in both A and B is clearly faster than the maximal rate of the bc_1 complex at steady-state with the same concentration of Q_1H_2 (50 s^{-1} , Fig. 6). Therefore, the rate in A and that in B can be considered to correspond to k_1 and k_4 , respectively, in the scheme of Fig. 4 for the steady-state mechanism of the enzyme.

of cytochrome c_1 with cytochrome *c* are maintained within the bc_1 complex (29, 43), the redox equilibrium between the two cytochromes can be simplified as $K_{\text{eq}} c = k_5/k_8$ within the scheme of Fig. 4. It can also be assumed that the binding constant of oxidized cytochrome *c* for the oxidized enzyme roughly corresponds to $K_d c_{\text{ox}} = k_6/k_5$ within the scheme of Fig. 4, since the binding properties of this substrate do not differ much if cytochrome c_1 is oxidized or reduced (20). Hence, the overall reaction of cytochrome *c* with its site can be approximated by the equivalences

$$k_5 k_7 / k_6 k_8 = K_i c_{\text{red}} / K_d c_{\text{ox}} = K_{\text{eq}} c \times k_7 / k_6. \quad [6]$$

Under the present assay conditions we measured a value of 1.65 for $K_{\text{eq}} c$ (as horse heart cytochrome *c* had a midpoint potential 13 mV more positive than beef cytochrome c_1), of 4 μM for $K_i c_{\text{red}}$ (Fig. 3 cf. Ref. (20)) and of 0.8 μM for $K_d c_{\text{ox}}$. By inserting such values in the relations of Eq. [6], k_7/k_6 becomes 3 and the factor $k_7 + k_6/k_7$ equals to 1.33. This latter number is particularly useful since, once inserted into Eq. [5], allows a more reliable estimation of the second-order rate constant k_5 , according to the equation

$$k_5 = K_{\text{min}} c \times 1.33. \quad [7]$$

From this it is also possible to deduce a minimal value of the k_7 rate constant after rearranging the above relations into the equation

$$k_7 = k_5 \times K_i c_{\text{red}} / K_{\text{eq}} c. \quad [8]$$

Such an evaluation of k_7 from manipulations of the steady-state data, although via a series of assumptions and approximations, avoids the difficulties of measuring the true rate of dissociation of reduced cytochrome *c* from the oxidized bc_1 complex. Previous measurements of this rate in the ascorbate-reduced enzyme apparently were incompatible with the TN_{max} obtained at steady state under the same conditions (20). However, this was largely because in the ascorbate-reduced enzyme the Rieske cluster acts as redox buffer on the reduction level of the cytochromes, thus lowering their oxidation rate (8, 9). Moreover, the approximate knowledge of k_7 from Eq. [8] is particularly valuable for deducing the kinetic parameters at the catalytic site for quinols, like ubiquinol-2, that are too hydrophobic to be used in pre-steady-state measurements.

3. Characterization of the Quinol Catalytic Site

Ubiquinol-1 is useful for characterizing the kinetic behavior of the Q catalytic site in the membrane domain of the bc_1 complex as: (i) its maximal turnover at steady state is clearly lower than that obtained with Q_2H_2 or DBQH_2 , thereby indicating that its kinetic interaction is more rate limiting than that of cytochrome *c*; (ii) it is sufficiently water soluble to allow the measurement of pre-steady-state rate constants that are kinetically competent with respect to the enzymatic turnover (10).

In principle, there are two quinol/quinone reaction sites in the membrane domain of the bc_1 complex, namely center *o* and *i* according to the Q-cycle terminology (11–16). The reaction at center *i* can easily reach equilibrium, particularly if the other center is blocked, e.g., by reduction of the Rieske cluster with ascorbate (4, 6, 9, 10, 14, 16). We have followed such an approach for measuring pre-steady-state reactions of Q_1H_2 and Q_1 with cytochrome *b* that are typical of center *i* (in that they are inhibited by

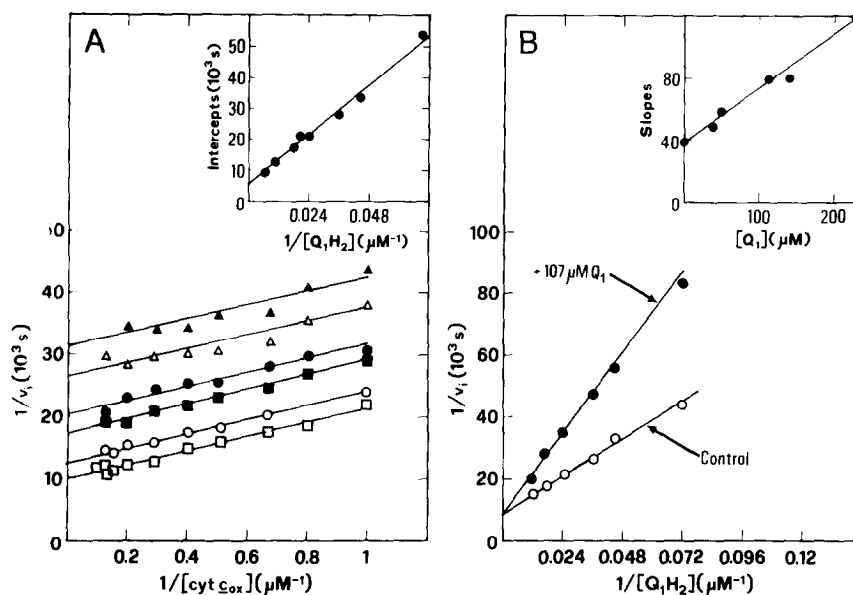


FIG. 6. Titrations of the ubiquinol-1:cytochrome *c* reductase in the beef *bc*₁ complex. The reductase assays have been performed with the same enzyme and quinol preparations as those used in the experiments of Fig. 5. (A) Cytochrome *c* titrations at the following fixed concentrations of Q₁H₂: (▲) 27 μM; (△) 33 μM; (●) 50 μM; (■) 63 μM; (○) 100 μM; (□) 162 μM. The insert shows the replot of the intercepts of the cytochrome *c* titrations (some of which are not shown for sake of clarity in the main graph) as a function of the concentration of Q₁H₂. The quinol titrations are also essentially parallel, even if they tend to cross each other at the highest concentrations of cytochrome *c* as in Ref. (19). It is to be noted that different batches of Q₁H₂ display quite different apparent *K_m* values, from 20 to 120 μM, probably due to a particular light sensitivity of this quinol (19, 20, 52). The average values of four separate Q₁H₂ titrations are presented in Table I to compare with those extrapolated in this assay. (B) Product inhibition of Q₁ versus Q₁H₂ with 5 μM cytochrome *c*. The insert shows the Dixon plot of the slope effect of Q₁, from which a *K_i*Q₁ of 96 μM is calculated according to Ref. (36).

antimycin (2–6, 9)) and could be equivalent to *k*₁ and *k*₄ in the reaction mechanism of Fig. 4 by assuming a rapid equilibrium at the Q site (Fig. 5). The equilibrium constant *K*_{eq}Q = *k*₁/*k*₄ evaluated from such measurements is 0.75, in agreement with the fact that the *E_m* of Q₁H₂ is more positive than that of cytochrome *b* and of the endogenous ubiquinone-10, which are the final electron acceptors in the enzyme at center *i* (10, 12, 16, 18, 45). Experiments with the fully oxidized enzyme showed a rate of cytochrome *b* reduction by excess of ubiquinol-1 close to that measured in the ascorbate-reduced complex as in (10), thereby suggesting that *k*₁ in the scheme of Fig. 4 can equally describe oxidation of ubiquinol at either center *i* or *o*. Moreover, the other 2,3-dimethoxy,5-methyl-6-substituted quinols used here produce an extent of reduction of cytochrome *b* at equilibrium which is similar to that seen with Q₁H₂ (ca. 30%). This allows one to assume with confidence that the *K*_{eq}Q for Q₁H₂ derived by transient kinetic measurements (Fig. 5) applies also for all these quinols, in accordance with Ref. (46).

Given this, we have applied the same treatment described before for the cytochrome *c* site to the reaction at the Q site within the scheme in Fig. 4. Although the binding of ubiquinols cannot be measured, that of ubiquinones, the products of the reductase catalysis, can be obtained from the independent approaches of fluorescence quenching (33) and product inhibition of the steady-state

activity (Fig. 3A). The latter is preferred in the case of ubiquinone-1, since this homolog does not fill completely the quinone binding site in the complex as deduced from its limited quenching of the intrinsic tryptophan fluorescence (33). The value of *K_i*Q₁ determined by product inhibition versus Q₁H₂ (96 μM, Fig. 6B) and the value of *k*₄, the rate constant of cytochrome *b* oxidation by Q₁ obtained from transient kinetics (2.6 × 10⁶ M⁻¹ s⁻¹, Fig. 5B), can be utilized for estimating *k*₃ according to the formula derived from the properties of the scheme in Fig. 4:

$$k_3 = K_i Q_1 \times k_4 = k_3/k_4 \times k_4 = 248 \text{ s}^{-1}. \quad [9]$$

By inserting such a value of *k*₃ in Eq. [1] and using the TN_{max} in the steady-state assays (Fig. 6), *k*₇ is estimated to be 727 s⁻¹, which is in reasonable agreement with the value of 928 s⁻¹ derived from Eq. [8]. Thus, the rapid-equilibrium assumptions for both the cytochrome *c* and the quinol catalytic site seem to be fundamentally correct. Now, from a rearrangement of Eq. [4], *k*₂ can be evaluated according to the equation

$$k_2 = k_3 (k_1/k_{\min} QH_2 - 1). \quad [10]$$

In the absence of pre-steady-state measurements, *k*₃ can

TABLE I
Typical Kinetic Parameters for Various Quinols in the bc_1 Complex from Beef Heart^a

Quinol	TN _{max} (s ⁻¹)	K _m (μM)	k ₁ (M ⁻¹ s ⁻¹)	k ₂ (s ⁻¹)	k ₃ (s ⁻¹)	K _d QH ₂ (μM)	K _i Q (μM)	K _d Q ^b (μM)
Q ₀ H ₂	6	266	3.1 × 10 ^{4h}	2	7	58	—	—
DDQH ₂	7	11	7.3 × 10 ^{5h}	2	8	2.7	—	—
Q ₁ H ₂ ^c	—	—	1.9 × 10 ⁶	—	248	—	—	—
Q ₁ H ₂ ^d	185	114	1.8 × 10 ⁶	23	226	13.2	96	11
Q ₁ H ₂ ^e	220	73	3.1 × 10 ⁶	75	293	24.5	73	—
PBQH ₂	258	92	3.8 × 10 ⁶	263	732	69	146	—
NBQH ₂	668	2.5	3.8 × 10 ⁸	1746	4048	5.1	7.9	6
DBQH ₂	558	3.0	2.8 × 10 ⁸	2820	4000	5.5	7.0	6
Q ₂ H ₂ ^f	770	5.7	3.8 × 10 ⁸	8290	4530	21.8	8.9	6.7
Q ₃ H ₂	208	1.2	1.3 × 10 ⁸	61	286	0.5	1.6	1.7
DQH ₂ ^g	14	200	1.0 × 10 ^{5h}	8	31	120	—	—

^a Using less than 10 μM horse-heart cytochrome *c* to avoid substrate inhibition. The experimental values of TN_{max} and K_m are the average values in at least three experiments or from the experiment that yields parameters that are closest to the mean.

^b Measured by quenching of the intrinsic tryptophan fluorescence (33).

^c Using the pre-steady-state values of k₁ and k₄ determined in the experiment of Fig. 5.

^d From the steady-state experiment of Fig. 6 that utilized the same reagents as that in Fig. 5.

^e Mean values from four separate experiments with different samples of Q₁H₂.

^f From the data of the experiments in Figs. 2 and 3.

^g Tetramethylbenzoquinol (duroquinol), for which a K_{eq}Q = 1.5 has been utilized to account for the lower midpoint potential of this quinol (4, 10, 47).

^h Calculated by the approximate relation expressed by Eq. [12].

be indirectly evaluated by using Eq. [1] with the value of k₇ from Eq. [8]. After deducing k₃, the value of k₁ can also be estimated from the equation

$$k_1 = 0.75 \times k_3 / K_i Q. \quad [11]$$

As the values from the pre-steady-state measurements (Fig. 5) are in good quantitative agreement with those obtained by the above equations from the steady-state properties of Q₁H₂ (Fig. 6 and Table I), the equations [1], [10], and [11] have been utilized for deducing approximate values of the constants in the scheme of Fig. 4 for other quinols. The most water-soluble quinones, e.g., Q₀, have a K_iQ that is too high to be accurately measured. In these cases it has been assumed that the ratio k₃/k₂ + k₃ is equal to the mean value of 0.73 obtained for the other ubiquinols, so that k₁ can be roughly estimated from the steady-state titrations according to the simplified form of Eq. [4]:

$$k_1 = 1.36 k_{\min} QH_2. \quad [12]$$

Table I summarizes all the kinetic parameters derived from the above equations in the beef heart enzyme under optimal assay conditions. The good correspondence between the K_iQ values and the K_d values (33) of the most hydrophobic quinones (Table I) underscores the validity of the elaborations based on the reaction scheme of Fig. 4. The values of the reaction constants reported recently by Rich and Harper (47) are comparable to the k₁ values

in Table I only for the most hydrophilic quinols up to Q₁H₂, whereas they appear to be much underestimated for the more hydrophobic quinols, e.g., Q₃H₂.

The major message that arises from the data in Table I is that quinols having less or more than 9–10 carbons in their hydrophobic tail display much lower values for k₁, k₂, and k₃, thus confirming previous indications of such a steric optimum in the specificity of the quinols in the bc_1 complex (30, 46–48). Interestingly, the K_d and K_iQ of Q₂ in the bc_1 complex are within the range of the K_d of either Q₂ or physiological quinones for the Q_B site in bacterial reaction centers (49, 50).

Table II summarizes the values deduced for the ubiquinol-2-cytochrome *c* reductase in bc_1 complexes purified from beef, turkey, chicken, and yeast mitochondria by methods involving fractionation with bile salts (10, 22, 28). It is consistently observed that k_{min}QH₂ is the experimental parameter displaying the least variability in the Q₂H₂ titrations, whereas the estimated value of k₃ exhibits a small range of variations in the different species. Moreover, the yeast enzyme shows the slowest k₁ and the highest K_iQ (Table I and Refs. (51, 52)). These differences appear to be significant and could be related to the rather different sequence of cytochrome *b* in yeast and in Vertebrata (2, 51).

DISCUSSION

*Further interpretation of the substrate inhibition of cytochrome *c* on the ubiquinol:cytochrome *c* reductase.*

TABLE II

Properties of the Ubiquinol-2: Cytochrome *c* Reductase in Different Preparations of the Mitochondrial *bc*₁ Complex^a

Kinetic constant	Unit	Beef	Turkey	Chicken	Yeast
c site					
k_7	s ⁻¹	928	928 ^b	970 ^c	928 ^b
k_5	$\times 10^8$ M ⁻¹ s ⁻¹	3.8	3.2	4.5 ^c	2.1 ^d
Experimental					
TN _{max}	s ⁻¹	770	794	721	792
K_m Q ₂ H ₂	$\times 10^{-6}$ M	5.7	4.8	3.8	7.9 ^d
k_{min} Q ₂ H ₂	$\times 10^8$ M ⁻¹ s ⁻¹	1.4	1.7	1.9	1.0 ^d
Q site					
k_3 ^e	s ⁻¹	4530	5498	3779	5404
k_2 ^f	s ⁻¹	8290	9486	5629	5548
k_1 ^g	$\times 10^8$ M ⁻¹ s ⁻¹	3.8	4.6	4.7	2.0
K_d Q ₂ H ₂	$\times 10^{-6}$ M	21.8	20.6	12.0	27.5

^a With horse heart cytochrome *c* at less than 15 μ M to prevent large substrate inhibition (22). The values for the beef heart refer to the experiments of Figs. 2 and 3, and are very close to the average of seven separate two-substrate titrations.

^b Considered to be equal to the value in beef owing to the very similar TN_{max} and K_m for cytochrome *c* (see also Refs. (22, 52)).

^c Calculated by Eqs. [7] and [8] using the experimental value of 3.17 μ M for K_{red} , 1.47 for K_{eq} and 3.4×10^8 M⁻¹ s⁻¹ for the typical k_{min} (cf. Ref. (22)).

^d Average value measured in four different wild type strains of *Saccharomyces cerevisiae* and in the baker's yeast enzyme (Ref. (51) and M. Degli Esposti and T. Tron, unpublished results).

^e Calculated from Eq. [1] using the tabulated values of k_7 .

^f Calculated from Eq. [10].

^g Calculated from Eq. [11] using $K_{eq}Q = 0.75$ in all cases and the K_1Q_2 values found experimentally in each enzyme: 8.9 μ M for both beef and turkey, 6 μ M for chicken, and 20 μ M (mean of three different yeast strains) for yeast (51).

An intriguing aspect of the steady-state activity of cytochrome *c* reductase is represented by the substrate inhibition exerted by cytochrome *c* (Fig. 2B and Refs. (19–22)). This phenomenon is considered to arise from binding of the substrate to noncatalytic sites in the reductase, generally identified as the negatively charged phospholipids and detergents (e.g., cholate) that are associated with the enzyme preparations (20). Although the apparent competition of cytochrome *c* versus ubiquinol looks bizarre, as each substrate binds to a separate site, it could also be a consequence of the unspecific binding to the complex. In fact, high concentrations of cytochrome *c* may lead to a direct oxidation of the reduced Rieske cluster—which cannot be reoxidized anymore via the intramolecular electron transfer to cytochrome *c*₁, owing to the saturation of the latter by cytochrome *c*.

The reaction between isolated Rieske protein and cytochrome *c* proceeds at a rate of 2×10^5 M⁻¹ s⁻¹ under the conditions of the present reductase assay (26) and therefore can substantially contribute to the observed rate of cytochrome *c* reduction when its concentration is above 10^{-5} M. The decreased steady-state level of the reduced

Rieske cluster produced by this unphysiological reaction with cytochrome *c* implies that higher concentrations of ubiquinol are now required for maintaining the same level of reduced complex. Hence, the direct reaction between the Rieske protein and cytochrome *c* would increase the observed rates of the reductase while shifting the saturation by ubiquinol to higher concentrations; the same is true for a (partial) competitive substrate inhibition by cytochrome *c*.

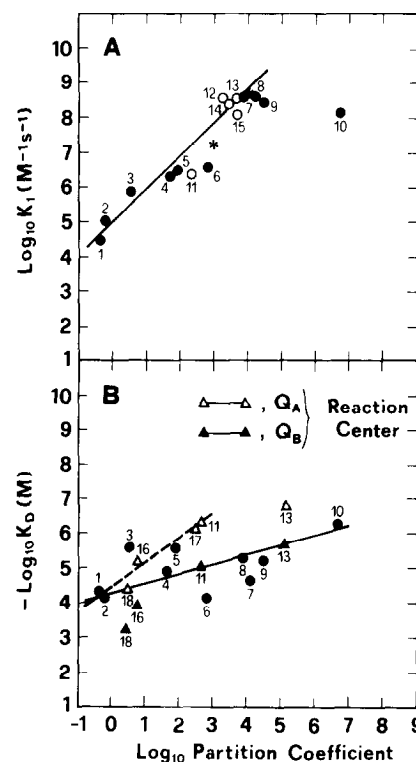


FIG. 7. Correlation between k_1 , K_d QH₂ and K_d Q with the hydrophobicity of various quinols and quinones. In both A and B the Log₁₀ of the partition coefficient (*P*) into cyclohexane/water are used as the basic reference for the hydrophobicity of the quinols and quinones. These *P* values have been taken from Ref. (47) or, when not available (e.g., NBQH₂), calculated from the value of the closest homolog and the offsets for the substituents reported in Ref. (53). (A) Correlation of the k_1 of the reductase activity for various quinols. (●) *P* in cyclohexane/water (47, 53) for the following quinols: 1, Q₀H₂; 2, DQH₂; 3, DDQH₂; 4, Q₁H₂ (k_1 from the data in Fig. 6); 5, Q₁H₂ (mean k_1 in Table I); 6, PBQH₂; 7, NBQH₂; 8, Q₂H₂; 9, DBQH₂; 10, Q₃H₂. (○) *P* in phospholipids/water (24) of the following quinones: 11, Q₁; 12, NBQ; 13, Q₂; 14, DBQ; and 15, Q₃. *, *P* in the mitochondrial membrane for Q₁H₂ and Q₂H₂ (nearly overlapping 7 and 8) calculated kinetically (25). Except for the last two cases, in which the original values of k_{1min} in Ref. (25) have been used, the k_1 values are taken from Table I. Note that if *P* in octanol/water (47, 50, 53) were used, the correlation with the k_1 values would be much less linear. (B) Correlation with the K_d for quinols (●, 1–15 as in A in the *bc*₁ complex (Table I) and comparison with the K_d of quinones for both the Q_A (Δ) and the Q_B (▲) site in the bacterial reaction center (data taken from Ref. (50)). 16, 17, and 18 identify the K_d for DDQ, DQ, and Q₀ in Ref. (50), respectively. All the other numbers identify the quinones as in A. The central line is a tentative fit to the overall set of K_d values, whereas the short dashed line on the left represents the linear fitting for the K_d Q_A values of the most hydrophilic quinols (50).

The Specificity of the Quinol Catalytic Site

This work provides the most accurate evaluation of the reaction rates of quinols at their site because the assays have been performed under conditions that minimize the distorting effect of high cytochrome *c* concentrations upon the kinetic properties of ubiquinol. As quinols and quinones should reach their catalytic site in membrane enzymes through the lipid phase of the bilayer (24, 25, 47, 50, 53), their lipid solubility is an important property affecting the kinetic specificity of the Q site in the bc_1 complex (24, 25, 33, 47). Herein, it is found that quinols' hydrophobicity expressed by their partition coefficients in cyclohexane/water shows a reasonable correlation with their k_1 values in the ubiquinol:cytochrome *c* reductase (Table I) up to Q_2H_2 (Fig. 7A). However, partition into cyclohexane as calculated from the data in Ref. (47, 53) seems to largely overestimate the hydrophobicity of $DBQH_2$ and Q_3H_2 relative to their k_1 values (Fig. 7A). A better correlation of these quinols is seen if the partition coefficients into lipids (24, 25) are considered (Fig. 7A). There is a clear correspondence, moreover, between the plateau in the apparent partition coefficients into lipids measured experimentally (24) and the apparent saturation of k_1 values of the quinols (Tables I, II and Fig. 7A). A parallel behavior is seen in the reconstitution of both Q_A and Q_B by exogenous ubiquinones in bacterial reaction centers (50). This suggests that a hydrophobic tail of 2 isoprenoid units possesses properties equivalent to those of the endogenous ubiquinol or ubiquinone with respect to both the kinetic and the physical interaction in the membrane reaction sites from water.

The hydrophobicity relationship of the K_d for the quinols is clearly different from that displayed by their k_1 , basically due to the large variations of the k_2 values (Table I and Fig. 7B). However, a tentative linear fit of the much scattered K_dQH_2 values as a function of the partition coefficients in cyclohexane/water seems to indicate that the affinity of the most hydrophobic quinols in the bc_1 complex is close to that displayed by their oxidized forms for the Q_B site in the reaction center (Fig. 7B, cf. Refs. (50, 54)).

The last observation has important implications for the mechanism of quinol reaction in the bc_1 complex, since it is consistent with structural similarities in the quinone-binding sites of cytochrome *b* and of the bacterial reaction centers (55) and with a functional similarity of center *i* with the Q_A – Q_B site in the reaction center (12, 55). Indeed, the electron input into the yeast bc_1 complex is best simulated if a permanently bound (endogenous) Q at center *i* mediates the reaction with exogenous quinol in a manner analogous to the reaction between Q_A and Q_B in the bacterial reaction center (10, 12, 50). This can be extended to the steady-state function of the enzyme in view of the kinetic competence of the pre-steady-state reactions at center *i* and of the rapid equilibrium attained at the Q site (Figs. 4–6, see also (10, 12)). Such an interpretation,

called “ Q_A -cycle” (12), also has the benefit of rationalizing some data of transient kinetics which apparently are contradictory to a general Q-cycle scheme, such as the faster reactivity of water-soluble quinols from the matrix side of the mitochondrial membrane (19, 56)—where center *i* is located (2)—and the sensitivity of electron input into the complex by the center *i* inhibitor, antimycin (4, 6, 10, 30, 56–58).

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